FAQ

1. How to enable loop detection in GPON ONU (SFU)

Switch_config_gpon0/2:1#gpon onu uni 1 loopback-detect enable

To configure the loopback on all ONU-

Switch#config
Switch_config#interface range gpON 0/1:1-10
Switch_config_if_range#gpon onu loopback-detect protocol private
Switch_config_if_range#gpon onu uni 1 loopback-detect enable.

Alternative option by making a profile-

Switch#config
Switch_config#loopback-detection

Need to make a profile

Switch_config#gpon profile onu-loopback-detection ***loop***
Switch-gpon-profile#gpon-profile port-vlan-table uni 1

Need to apply that profile on the ONU

Swithc_config#interface gpon0/2:1
Switch_config_gpon0/2:1#gpon onu loopback-detect protocol ctc
Switch_config_gpon0/2:1#gpon profile onu-loopback-detection ***loop***

2. How to enable the "telnet" in BDCOM Switch please use the following commands in config mode -

Switch#config
Switch_config#aaa authentication login default local
Switch_config#aaa authentication enable default none
Switch_config#aaa authorization exec default local

3. How to increasing the session time in OLT

Switch_config#line ?

console -- Primary terminal line

vty -- Virtual terminal

Switch_config#line console 0

Switch_config_line#exec-timeout ?

<0-86400>

Switch_config_line#exec-timeout

NB: 0 (zero) means no time out

4. How to configure back-up link in Switch

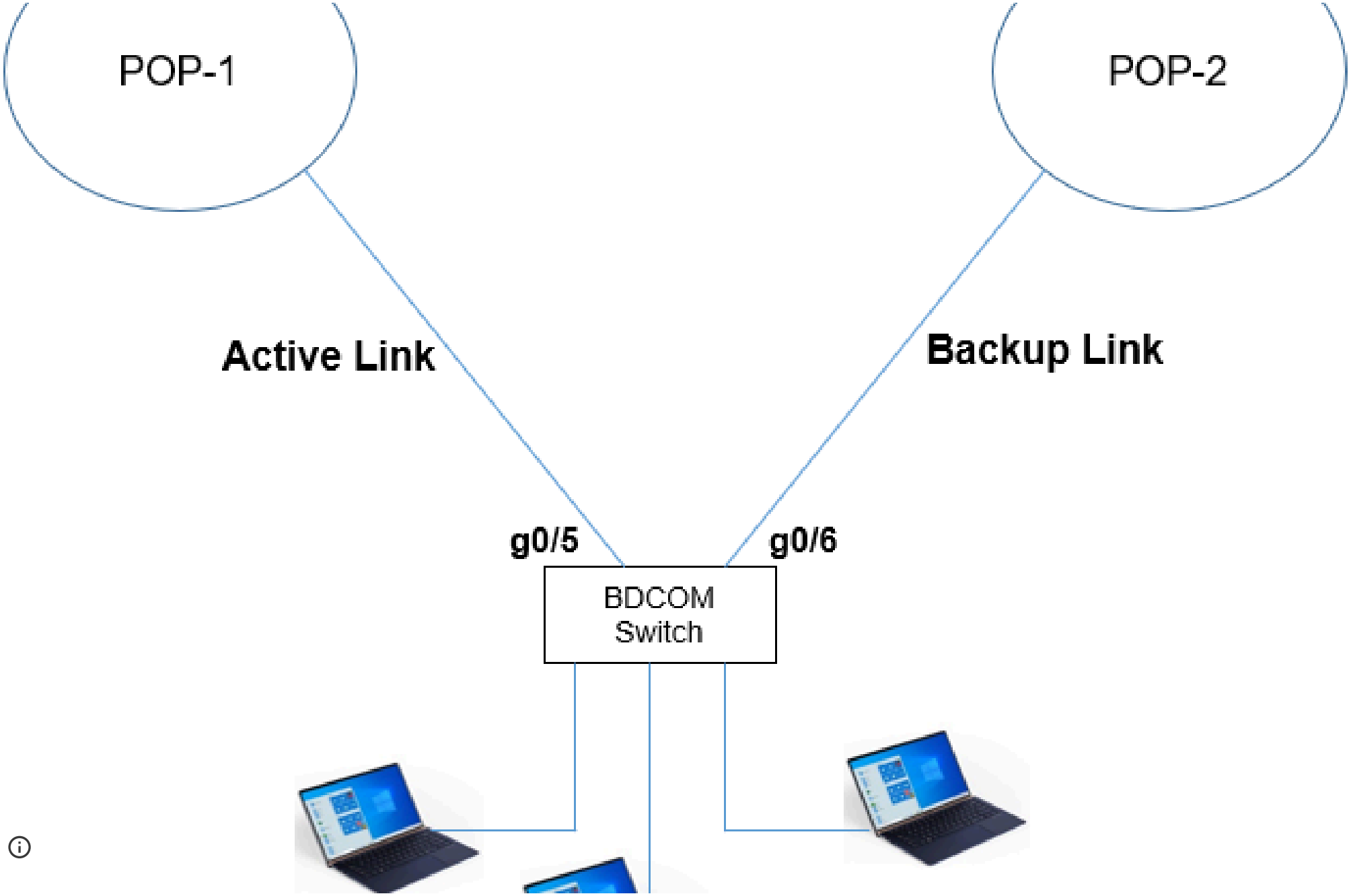
Switch_config_g0/5#backup-link-group 1 active

Switch_config_g0/5#exit
Switch_config#interface GigaEthernet 0/6
Switch_config_g0/6#backup-link-group 1 backup

NB: Here, if the active link fails then the traffic will pass through Backup link automatically. Use the following command to shift the Backup link again to Active link automatically

Switch_config#backup-link-group 1 preemption-mode forced delay 0

To check the backup link, use below command
#show backup-link-group 1


POP-1
POP-2
Active Link
Backup Link
g0/5
g0/6
BDCOM
Switch

5. How to delete ONU from CLI mode

For EPON

```
Switch#config
Switch_config#interface epon 0/1
Switch_config_epon0/1#no epon bind-onu sequence ?
    <1-64>    -- LLID sequenceD
```

For GPON

```
Switch#config
Switch_config#interface gpON 0/1
Switch_config_gpon0/1#no gpon bind-onu sequence ?
  WORD    -- ONU sequenceIDs such as (1,3-5,7), max 128
```

6. How to bind ONU with OLT from CLI mode

```
Switch#config
Switch_config#interface epon 0/3
Switch_config_epon0/3#epon bind-onu mac xxxx.xxxx.xxxx
Switch_config_epon0/3#exit
Switch_config#wr all
```

7. How to set an IP address to ONU (SFU) from EPON OLT

Switch_config_epon0/5#switchport trunk vlan-untagged none
Switch_config_epon0/5#exit
Switch_config#

Switch_config#interface epON 0/5:6
Switch_config_epon0/5:6#epon onu ctc ip address static 172.16.0.100 255.255.255.0 gateway 172.16.0.1 cvlan 1 svlan 0 priority 5

NB: Here, 172.16.0.100 is for ONU IP and gateway 172.16.0.1 is OLT IP

For different VLAN

Switch_config#interface epON 0/5:6
Switch_config_epon0/5:6#epon onu port 1 ctc vlan mode trunk 100 1518
Switch_config_epon0/5:6#epon onu ctc ip address static 172.16.0.100 255.255.255.0 gateway 172.16.0.1 cvlan 1518 svlan 0 priority 5

8. How to configure SNMP in OLT

Switch_config#snmp-server community 0 <***BDCOM***> RW

Switch_config#snmp-server host <172.16.2.30> version v2c <***BDCOM***> authentication configure snmp

NB: Here "**BDCOM**" is the community name and the IP **172.16.2.30** is your PC IP

9. How to set an IP address in ONT from GPON OLT /Switch

Switch#config

Switch_config#inteface gpon0/2:1

Switch_config_gpon0/2:1#gpon onu ip-host ?

<1-1> -- ONU ip-host port ID

Switch_config_gpon0/2:1#gpon onu ip-host 1 address static 172.16.0.100 255.255.255.0 172.16.0.1

10. How to configure VLAN in SFU ONU in CLI mode

Switch_config#interface epON 0/5:1
Switch_config_epon0/5:1#epon onu port 1 ctc vlan mode tag 200 priority 0 **[Here the example VLAN is 200]**

For GPON ONU

Step01: To create a profile

Switch#config
Switch_config#gpon profile onu-vlan ***V100*** **[Here V100 is the profile name]**
Switch-gpon-profile#gpon-profile vlan mode tag
Switch-gpon-profile#gpon-profile vlan pvid 100

Step02: Apply the profile on an ONU

Switch_config#interface gpON 0/1:1
Switch_config_gpon0/1:1#gpon onu uni 1 vlan-profile ***V100*** **[Here V100 is the profile name]**

11. How to do port level redundancy in GPON using with 2:N Splitter

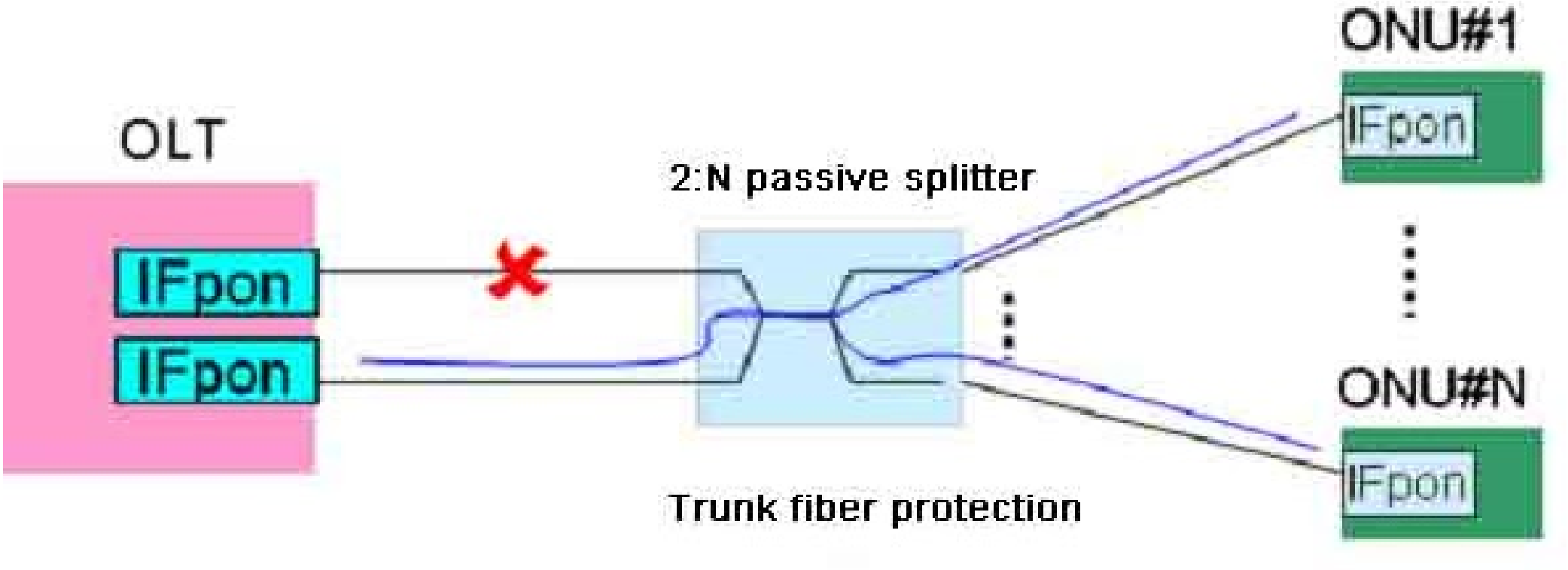
Switch_config#interface gpsG 0/1

Switch_config_gpsg0/1#gpon psg member active gPON 0/1 standby gPON 0/2

Switch_config_gpsg0/1#no shut

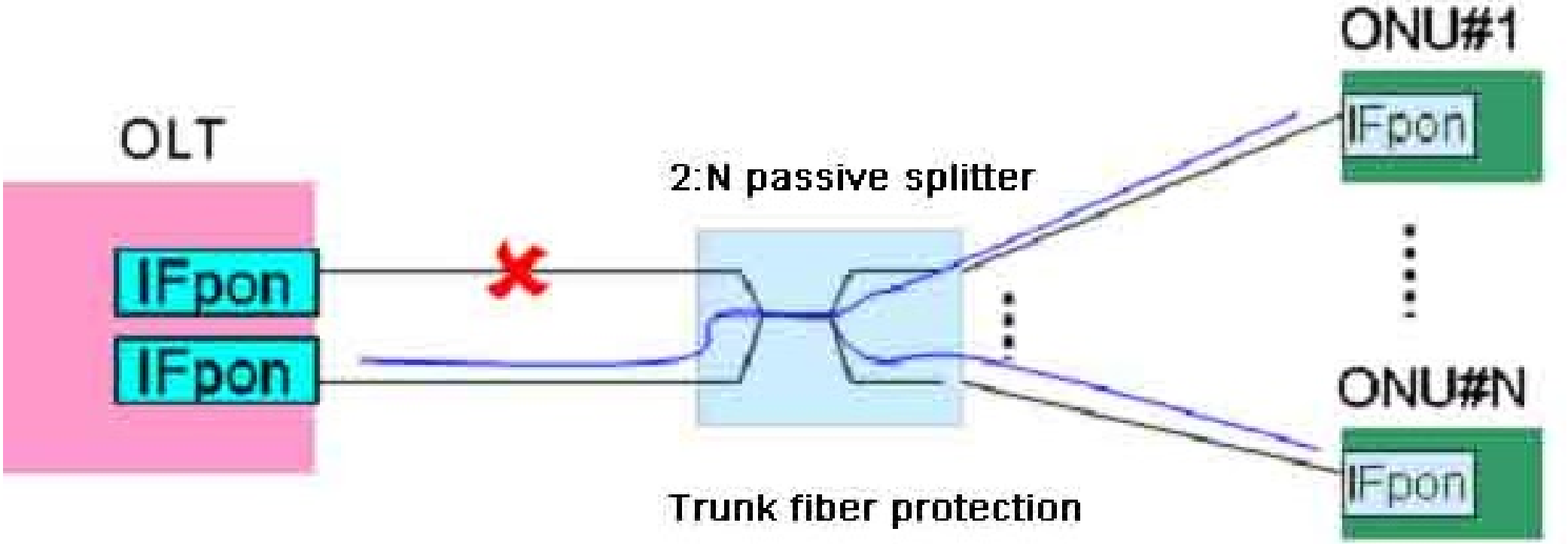
Switch_config_gpsg0/1#exit

Switch_config#exit



12. How to do port level redundancy in EPON using with 2:N Splitter

```
Switch_config_psg0/1#epon psg member active ePON 0/1
Switch_config_psg0/1#epon psg member standby ePON 0/2
Switch_config_gpsg0/1#no shut
Switch_config_gpsg0/1#exit
Switch_config#exit
Switch#wr all
```



13. How to control storm at EPON ONU

Switch_config_epon0/1:1#epon onu port 1 storm-control mode ?

1 -- limit broadcast

2 -- limit multicast

3 -- limit unknown unicast

4 -- limit all packet

Switch_config_epon0/1:1#epon onu port 1 storm-control mode 1 threshold ?

<256-100000> -- Enter part of storm suppression level(unit:kbps)

Switch_config_epon0/1:1#epon onu port 1 storm-control mode 1 threshold

14. How to enable loop detection in ONU from OLT

Switch_config#interface range epON 0/1:1-13

Switch_config_if_range#epon onu all-port ctc loopback detect

Switch_config_if_range#epon onu all-port ctc notify loopback

15. How to set enable password

Switch_config#enable password 0 <password>

Switch_config#aaa authentication enable default enable

16. How to change Native VLAN in Switch or OLT

```
Switch_config_g0/2#switchport mode trunk
Switch_config_g0/2#switchport trunk vlan-allowed 999,500,510
Switch_config_g0/2#switchport pvid 999
Switch_config_g0/2#exit
Switch_config#wr all
```

OUTPUT

!

interface GigaEthernet0/2

switchport trunk vlan-allowed 500,510,999

switchport mode trunk

switchport pvid 999 **[Here Native VLAN is 999]**

17. How to change lost time of EPON ONU ?

How to Set aging time for ONU SN binding entry when ONU auto unbind, in inactive or offline?

Switch#config
Switch_config#epon auto-clear-onu-lost enable
Switch_config#epon onu-lost-time ?
<10-10000000> -- onu lost time value (in seconds, default is 2592000 (30days))

Switch_config#epon onu-lost-time ***604800*** [In seconds, set 7days Value]
Switch_config#exit
Switch#clear epon lost-onu
Switch#write all

For GPON

Switch#config
Switch_config#gpon onu-auto-unbind ?
<0, 1000-1000000000> -- aging time (0 default), 0 never (sec)

Switch_config#gpon onu-auto-unbind ***604800*** [In seconds, set 7days Value]
Switch_config#write all

18. How to create a limited user in OLT or Switch with a limited privilege

Switch_config_<group name>#exec privilege default 8

Switch_config#username <username> password 0 <password> author-group <group name>

Example -

Switch_config#localauthor NOC
Switch_config_localauthor_NOC#exec privilege default 8
Switch_config#username test password 0 test321 author-group NOC

Switch#config
Switch_config#aaa authorization exec default local

Switch_config#localauthor ***support***
Switch_config_localauthor_support#exec privilege default 5

Switch_config#username ***test*** password 0 **test** author-group ***support***
Switch_config#privilege exec 5 show run

19. How to disable EPON ONU

Switch#config

Switch_config#interface epON 0/1

Switch_config_epon0/1#epon onu-blacklist mac xxxx.xxxx.xxxx

To check the status -

Switch#show epon interface epON 0/1:1 onu ctc optical-transceiver-diagnosis

20. How to disable GPON ONU

Switch_config_gpon0/1:1#exit

To check the status -

Switch#show gpon interface gpON 0/1:1 onu optical-transceiver-diagnosis

21. How to check optical power (RX) of EPON and GPON ONU

For EPON

#show epon interface epON 0/1:1 onu ctc optical-transceiver-diagnosis
#show epon onu-ctc-optical-transceiver-diagnosis interface epON 0/1

NB: These commands are to check ONU RX from client side

#show epon optical-transceiver-diagnosis interface epON 0/1

NB: This command is to check ONU RX from OLT side

For GPON

#show gpon interface gpON 0/1:1 onu optical-transceiver-diagnosis
#show gpon onu-optical-transceiver-diagnosis interface gpON 0/1

NB: These commands are to check ONU RX from client side

#show gpon optical-transceiver-diagnosis interface gpON 0/1

NB: This command is to check ONU RX from OLT side

22. How to reboot ONU

Switch#epon reboot onu interface epON 0/1:1

For GPON ONT

Switch#gpon reboot onu interface gpON 0/1:1

23. How to shutdown ONU LAN port

For EPON

Switch#
Switch#config
Switch_config#interface epON 0/1:2
Switch_config_epon0/1:2#epon onu port 1 ctc shutdown

For GPON

Switch#
Switch#config
Switch_config_gpon0/1:2#
Switch_config_gpon0/1:2#gpon onu uni 1 shutdown

24. How to control storm in EPON/GPON/Switch

Switch_config_g0/1#storm-control ?

broadcast -- Broadcast address storm control

multicast -- Multicast address storm control

unicast -- Unknown-Unicast address storm control

Switch_config_g0/1#storm-control broadcast threshold ?

<1-1048575> -- Enter Integer part of storm suppression level(PPS)

25. How to check the LAN port status of ONU/ONT (only for SFU)

For EPON ONU

Switch#show epon interface epON 0/1:1 onu port 1 state
Hardware state is Link-Up
Speed is 1000Mbps
Duplex is Full-Duplex

For GPON ONT

Switch#

Switch#show gpon interface gpON 0/1:1 onu port 1 state
GPON0/1:1 uni-port 1 up
10/100/1000 BASE-T(1Gbps Full-Duplex)

26. How to set port security by MAC in ONU

Switch_config_epon0/1:2#switchport port-security mode dynamic
Switch_config_epon0/1:2#switchport port-security dynamic maximum ?
<1-1021> -- Configure interface maximum address

27. How to increase the ONU bandwidth capacity to 1Gbps from OLT

Switch#config

Switch_config#interface epON 0/1:1

Switch_config_epon0/1:1#epon sla downstream pir ?

<512-1000000> -- Enter peak bandwidth in units of kbps

Switch_config_epon0/1:1#epon sla downstream pir 1000000 cir ?

<1-960000> -- Enter committed bandwidth in units of kbps

Switch_config_epon0/1:1#epon sla downstream pir 1000000 cir 960000 fir ?

<0-950000> -- Enter fixed bandwidth in units of kbps

28. How to see MAC address table in ONU/ONT

Mac Address Table (Total 1)

Vlan Mac Address Type Ports

---- ---------- ---- -----

1 **1062.eb21.52b8** DYNAMIC epon0/1:2

Switch#show mac address-table interface gpON 0/1:1

Mac Address Table (Total 1)

Vlan Mac Address Type Ports

---- ---------- -------- -------

350 **c025.e9aa.6c79** DYNAMIC gpon0/1:1-1

29. How to set ONU/ONT authentication in GPON/EPON OLT with SN or MAC

To set the authentication method

Switch#config

Switch_config_gpon0/1#gpon onu-authen-method sn

To check the rejected ONT

Switch#show gpon onu-rejected-information

To bind the ONT with SN authentication

Switch#config
Switch_config#interface gpON 0/1

Switch_config_gpon0/1#gpon bind-onu sn ?

```
XXXXXXXXXXXXXXXX   -- SN(16 characters)
XXXX:XXXXXXXX      -- SN(VendorID:SerialNumberOfVendor)
```

For EPON

Switch#

Switch_config#

Switch_config#interface epON 0/1

Switch_config_epon0/1#epon onu-authen-method mac

To bind the ONT with SN authentication

Switch_config_epon0/1#epon bind-onu mac ?

```
<xxxx.xxxx.xxxx>     -- MAC address Value
```

30. How to enable SSH in OLT (EPON/GPON) and Switch

Switch_config#ip sshd enable

31. How to configure DHCP snooping

Switch#config
Switch_config#ip dhcp-relay snooping
Switch_config#ip dhcp-relay snooping vlan 1
Switch_config#ip arp inspection vlan 1
Switch_config#ip verify source vlan 1

To trust the DHCP server

Switch#config
Switch_config#interface gigaEthernet 0/1 **[The trusted server is connected from gig 0/1]**
Switch_config_g0/1#dhcp snooping trust
Switch_config_g0/1#arp inspection trust
Switch_config_g0/1#ip-source trust

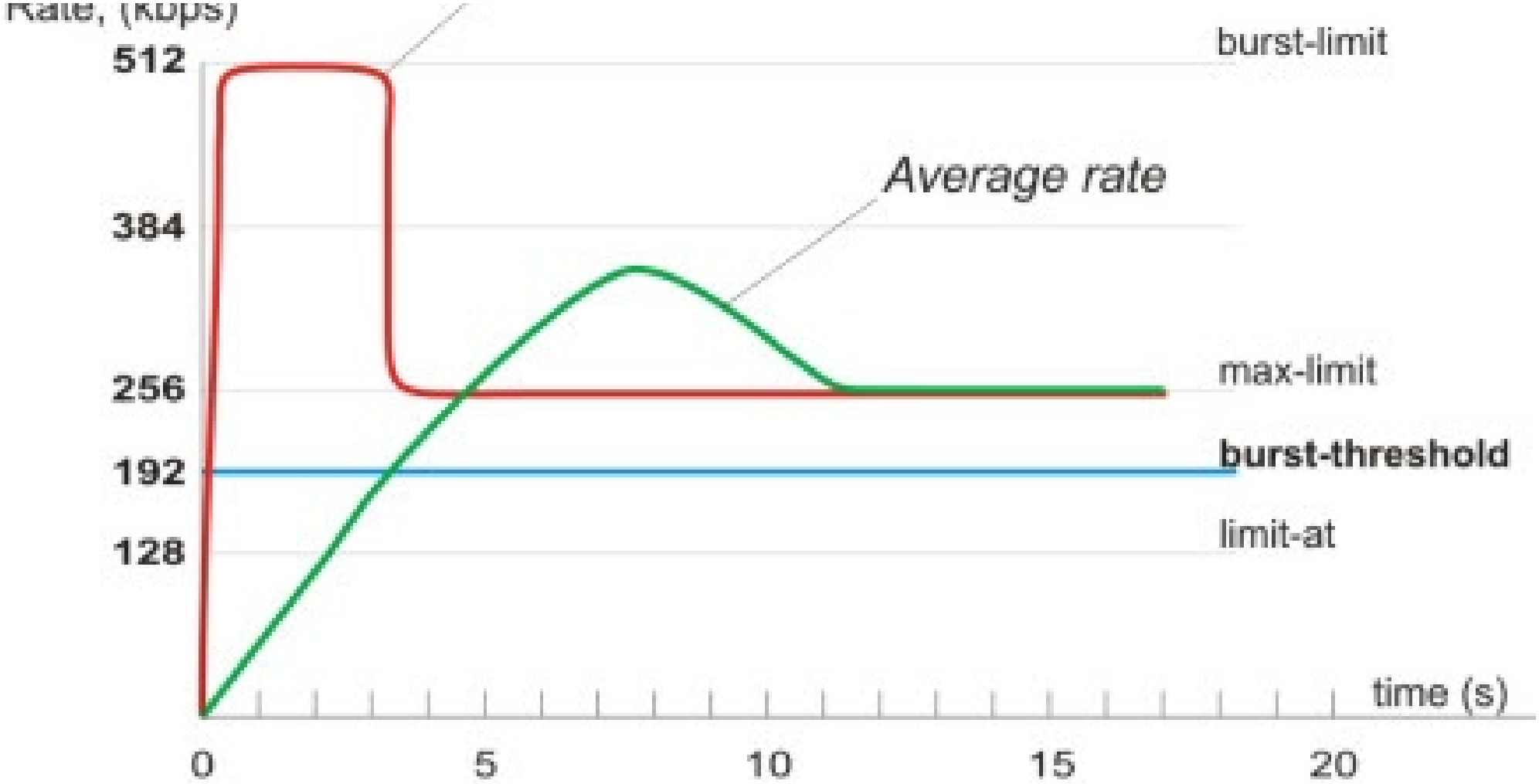
32. How to set rate-limit in EPON ONU (Assign bandwidth in ONU LAN port)

```
Switch_config_epon0/1:1#epon onu port 1 ctc rate-limit ?

<128-1000000>                -- Configure Bandwidth(unit:kbps)

Switch_config_epon0/1:1#epon onu port 1 ctc rate-limit 100000 ?

  ingress  -- Configure port rate-limit ingress
  egress   -- Configure port rate-limit egress
```



33. How to set rate-limit in GPON ONT (PON port / WAN port)

Create a profile. Let's say the profile name is ***1M***
Switch#config
Switch_config#gpon profile onu-tcont ***1M***

Step02:

Assign the bandwidth (CIR and PIR) on ***1M*** profile
Switch_config_prof_onuTcont_1M#gpon-profile tcont-type 3 pir ?

<512-1244160> -- Max Bandwidth(kbps)

Switch_config_prof_onuTcont_1M#gpon-profile tcont-type 3 pir 1024 cir ?

<256-768> -- Assured Bandwidth(kbps)

Switch_config_prof_onuTcont_1M#gpon-profile tcont-type 3 pir 1024 cir

<256-768> -- Assured Bandwidth(kbps)

Bind the virtual port. Let's say the virtual port name is ***T1M***
Switch_config#gpon profile onu-tcont-virtual-port-bind ***T1M***
Switch_config_prof_onuTcontVportBind_T25M#gpon-profile virtual-port 1 profile virtual-port-default tcont 1 profile ***1M***

Step04:

Apply the profile on a specific ONT (0/1:1) and rate limit
Switch_config_gpon0/1:1#gpon onu tcont-virtual-port-bind-profile ***T1M***
Switch_config_gpon0/1:1#gpon onu virtual-port 1 downstream rate-limit ?

<1-38880>

NB: For GP3600B version OLT needs to add this command in PON port

gpon qos-mode virtual-port egress to configure rate-limit

34. How to make DHCP server in OLT/Switch

Create a VLAN 100 and assign IP address

Switch#config

Switch_config#interface vlAN 100

Switch_config_v100#ip address 172.16.100.1 255.255.255.0

Create IP pool name, IP range and then enable DHCP server

Switch_config#ip dhcpd pool test
Switch_config_dhcp#network 172.16.100.0 255.255.255.0
Switch_config_dhcp#range 172.16.100.2 172.16.100.254
ip dhcpd enable

35. How to set rate limit (ingress and egress) to OLT or Switch

```
OLT#config
OLT_config#interface gpon 0/2
Switch_config_gpon0/2#switchport rate-limit ?
  <1-16383>   -- Configure Bandwidth (unit:64kbps)
   bandwidth  -- Configure Bandwidth(unit:1% of physical bandwidth)

OLT_config_gpon0/2#switchport rate-limit 1600
  ingress  -- Config port rate-limit ingress incoming traffic
  egress   -- Config port rate-limit egress outgoing traffic

OLT_config_gpon0/2#switchport rate-limit 1600 ingress
OLT_config_gpon0/2#switchport rate-limit 1600 egress
```

NB: Here, 1 meaning 64Kbps, 2 meaning 128Kbps, 10 meaning 640Kbps and 1600 meaning (1600 x 64) = 102,400Kbps which is 100Mbps

For Switch port

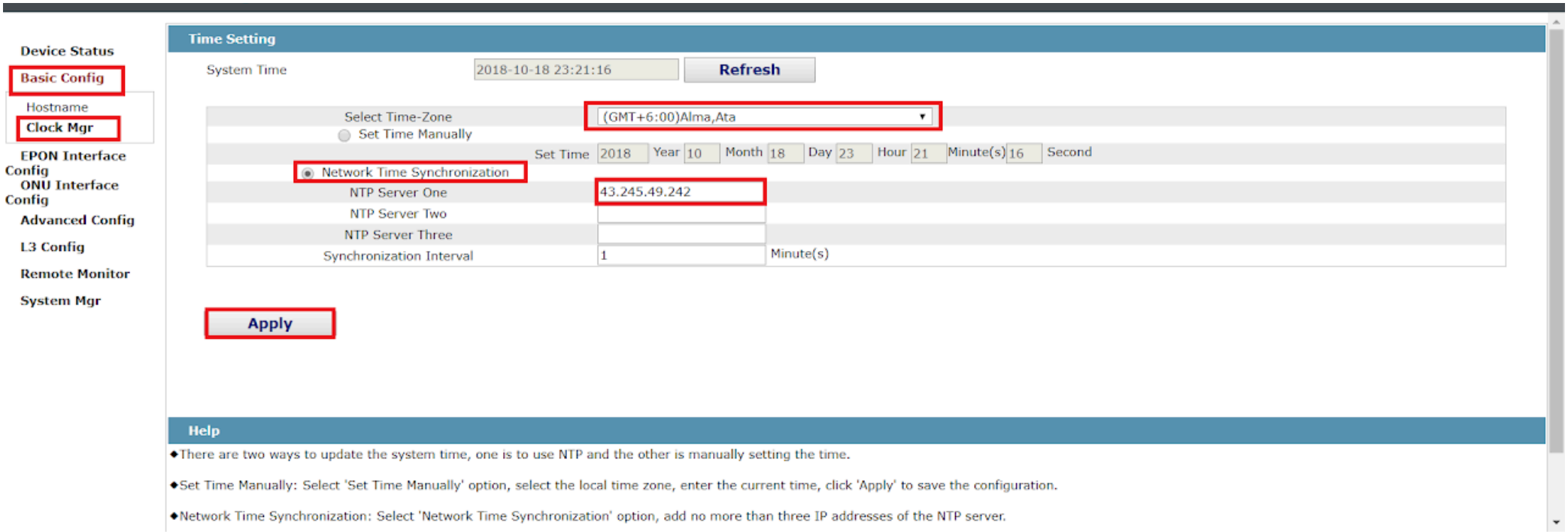
```
Switch#config
Switch_config#interface gig0/2
Switch_config_g0/1#switchport rate-limit ?
  <1-16383>   -- Configure Bandwidth(unit:64kbps)
  bandwidth  -- Configure Bandwidth(unit:1% of physical bandwidth)

Switch_config_g0/2#switchport rate-limit 1600
  ingress  -- Config port rate-limit ingress incoming traffic
  egress   -- Config port rate-limit egress outgoing traffic

Switch_config_g0/2#switchport rate-limit 1600 ingress
Switch_config_g0/2#switchport rate-limit 1600 egress
```

NB: Here, 1 meaning 64Kbps, 2 meaning 128Kbps, 10 meaning 640Kbps and 1600 meaning (1600 x 64) = 102,400Kbps which is 100Mbps

36. How to set NTP in OLT and Switch (For Bangladesh IP is 194.225.150.25)



37. How to implement policy-map ? (Assign bandwidth on VLAN)

Switch-policy-map#classify vlan 100
Switch-policy-map#action bandwidth ?
<1-163840> -- Configure Bandwidth (unit:64Kbps)
Switch-policy-map#action bandwidth 1600
Switch-policy-map#exit

NB: Here the name of policy map is ***BW_100Mb_V100*** and
1 meaning 64Kbps, 2 meaning 128Kbps, 10 meaning 640Kbps and 1600 meaning (1600 x 64) = 102,400Kbps which is 100Mbps

Switch_config#interface gigaEthernet 0/1

Switch_config_g0/1#qos policy ***BW_100Mb_V100*** ?

ingress -- Config port policy map ingress
egress -- Config port policy map egress

NB: Implement the policy map on VLAN 100 in the interface of gig0/1

38. How to configure TACACS+ and Radius

Switch_config#aaa authentication enable default none
Switch_config#aaa authorization commands 15 default group tacacs+
Switch_config#aaa authorization commands 0 default group tacacs+
Switch_config#aaa authorization exec default group tacacs+ local
Switch_config#aaa accounting commands 15 default start-stop group tacacs+
Switch_config#aaa accounting commands 0 default start-stop group tacacs+
Switch_config#aaa accounting network default start-stop group tacacs+
Switch_config#tacacs-server host ***172.25.102.61***
Switch_config#tacacs-server key ***abc123xyz***

For Radius

Switch_config#aaa authentication login default group radius
Switch_config#aaa authentication enable default none
Switch_config#aaa authorization commands 15 default group radius
Switch_config#aaa authorization commands 0 default group radius
Switch_config#aaa authorization exec default group radius local
Switch_config#aaa accounting commands 15 default start-stop group radius
Switch_config#aaa accounting commands 0 default start-stop group radius
Switch_config#aaa accounting network default start-stop group radius
Switch_config#radius-server host ***172.25.102.61***
Switch_config#radius-server key ***abc123xyz***

39. How to configure Jumbo frame in GPON ONT (Maximum size is 1996, default is 1518)

Switch#config

Switch_config#gpon profile onu-uni ***mtu_jumbo***

Switch_config_prof_onuEth_mtu_jumbo#gpon-profile max-frame-size 1996

Step 02: Apply the profile "mtu_jumbo" on an ONT

Switch_config# interface GPON0/1:1

Switch_config_gpon0/1:1#gpon onu uni 1 uni-profile ***mtu_jumbo***

40. How to enable data forwarding among ONUs under the same EPON OLT PON port

Switch#config

Switch_config#interface epON0/1

Switch_config_epon0/1#epon inner-onu-switch

41. How to forward data among the UNI ports of ONU in EPON

Switch#config

Switch_config#interface epON0/1:1

Switch_config_epon0/1:1#no epon onu port-protect

42. How to filter/block and forwared/allow MAC in GPON ONU

Switch#config
Switch_config#gpon profile onu-mac-filter ***Block***
Switch-gpon-profile#gpon-profile filtertype filter
Switch-gpon-profile#gpon-profile entry 1 address a08c.fd2a.7342

Step 02: Apply the profile "Block" on the specific ONU

Switch#config
Switch_config#interface GPON0/1:1
Switch_config_gpon0/1:1#gpon onu mac-filter-profile ***Block***

Step 01: Create a profile named "Allow"

Switch#config
Switch_config#gpon profile onu-mac-filter ***Allow***
Switch-gpon-profile#gpon-profile filtertype forward
Switch-gpon-profile#gpon-profile entry 1 address a08c.fd2a.7342

Step 02: Apply the profile "Allow" on the specific ONU

Switch#config
Switch_config#interface GPON0/1:1
Switch_config_gpon0/1:1#gpon onu mac-filter-profile ***Allow***

43. How to configure multicast in GPON HGU ONU

```
 gpon-profile entry 1 multicast-group-address-range 239.10.10.1
!
gpon profile onu-mcst-oper test id 2
 gpon-profile downstream-mcst-tag strip
!
!
interface g0/1
switchport mode trunk
!
interface gpon0/1
switchport mode trunk
!
interface GPON0/1:2
 gpon onu tcont-virtual-port-bind-profile tvbind-default
 gpon onu flow-mapping-profile flow-mapping-default
 gpon onu virtual-port 1 gem-port 258
 gpon onu veip 1 mcst-dynamic-group-profile test
 gpon onu veip 1 mcst-oper-profile test

!
ip mcst enable
ip mcst querier enable
ip igmp-proxy enable
ip mcst mrouter interface GigaEthernet0/1
ip mcst mc-vlan 1000 range 239.10.10.1
!
vlan 1000
```

44. How to check distance of the ONU from OLT

For EPON

#show epon active-onu

For GPON

#show gpon interface gpON 0/1:1 onu basic-info

Switch#config
Switch_config#gpon profile onu-tcont ***T20***
gpon-profile tcont-type 3 pir 20480 cir 20224
gpon profile onu-rate-limit ***R20***
gpon-profile pir 20480 cir 20480
gpon profile onu-virtual-port ***VP20***
gpon-profile upstream rate-limit-profile ***R20***
gpon profile onu-tcont-virtual-port-bind ***TV20***
gpon-profile virtual-port 1 profile VP20 tcont 1 profile ***T20***
gpon profile onu-flow-mapping ***F20***
gpon-profile entry 1 uni type eth-uni 1
gpon-profile entry 1 virtual-port 1

==========

Switch#config
Switch_config#gpon profile onu-vlan ***V100***
Switch-gpon-profile#gpon-profile vlan mode tag
Switch-gpon-profile#gpon-profile vlan pvid 100

Step 02: Make a template

Switch#config
Switch_config#gpon onutype-template ***SFU***
Switch_config_onutype_SFU#gpon-onutype match ctc-onu-type ***SFU***
Switch_config_onutype_SFU#gpon-onutype config tcont-virtual-port-bind-profile TV20
Switch_config_onutype_SFU#gpon-onutype config flow-mapping-profile F20
Switch_config_onutype_SFU#gpon-onutype config virtual-port 1 downstream rate-limit 320
Switch_config_onutype_SFU#gpon-onutype config loopback-detect protocol private
Switch_config_onutype_SFU#gpon-onutype config uni 1 vlan-profile ***V100***
Switch_config_onutype_SFU#gpon-onutype config uni 1 loopback-detect enable

Step 03: Apply the template on the PON port of the OLT

Switch#config
Switch_config#interface GPON0/1
S ı_config_gpon0/1# gpon bind-onutype ***SFU*** precedence 1

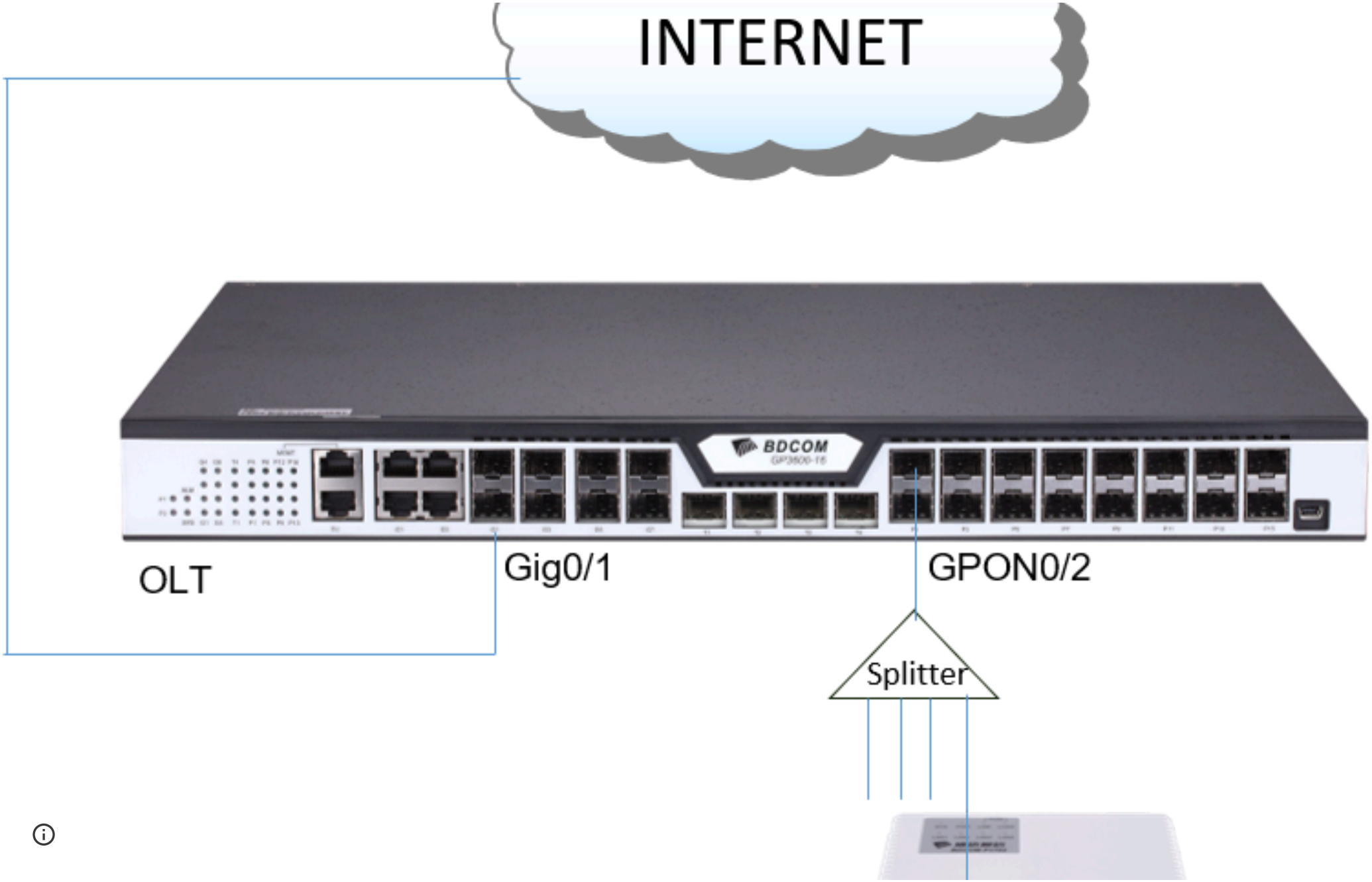
Switch_config# gpon profile onu-flow-mapping ***test***
Switch-gpon-profile#gpon-profile entry 1 uni type eth-uni all
Switch-gpon-profile#gpon-profile entry 1 virtual-port 1
Switch-gpon-profile#no gpon-profile entry 1 port-protected

Step02: Apply the profile on the ONU

Switch_config# interface GPON0/2:4
Switch_config_gpon0/2:4# gpon onu flow-mapping-profile ***test***

NB: If it is in different PON port need to use **"no switchport protected"** command and if it is in same PON port need to use "**switchport reflect-bridge"** command

This solution is applicable for only few models ONU with special firmware. That's why please contact with BDCOM team for this solution


INTERNET
BDCOM
OLT
Gig0/1
GPON0/2
Splitter

47. How to configure port mirror in Switch/OLT

```
Switch#config
Switch_config#mirror session 1 destination interface g0/1
Switch_config#mirror session 1 source interface g0/2 both
```

NB: Here source port is the testing port and destination port is the packet capturing port (where the laptop is connected with Wireshark)

48. How to reset Switch/OLT if you forget the password

1. Connect the console cable to the Switch/OLT with BDCOM console cable which is provided with the Switch/OLT.
2. Reboot the Switch/OLT by power off/on
3. When Switch/OLT is rebooting just press "**CTRL + P**"
4. You will be able to see **monitor#** mode in the screen
5. Please use the command "**show configuration**". You will be able to see the password in clear text if the password was not encrypted
6. If you want to reset (factory default) the Switch/OLT use the command "delete startup-configuration"
7. Reboot the Switch/OLT and you will be able to access with default mode. username and password will be admin/admin.
8. Your all configuration will be lost and you need to configure your Switch/OLT again

49. How to configure ACL (Access Control List) in BDCOM Switch/OLT

Step01: Make a access-list named test

Swith#config
Switch_config#ip access-list extended ***test***
Switch_config_ext#permit tcp any any eq 22
Switch_config_ext#permit tcp any any eq 161
Switch_config_ext# deny ip any any

Step02: Apply the access-group on the interface

Switch_config#
Switch_config#interface gigaEthernet 0/1
Switch_config_g0/1#ip access-group ***test***

B. Objective: Only from 192.168.100.1 to 192.168.100.254 IPs are allowed to browse the switch in http port at 80. Other all IP will be denied

Swith#config
Switch_config#ip access-list standard ***test***
Switch_config_std#permit 192.168.100.0 255.255.255.0
Switch_config_std#exit

Switch_config#ip http access-list ***test***

50. How to configure VOICE VLAN

Swith#config
Switch_config#interface GigaEthernet0/1
Switch_config_g0/1#description IP-Phone
Switch_config_g0/1#switchport trunk vlan-allowed 500,400
Switch_config_g0/1#switchport trunk vlan-untagged 500, 400
Switch_config_g0/1#switchport pvid 500
Switch_config_g0/1#switchport mode trunk
Switch_config_g0/1#switchport voice-vlan mode mac-address
Switch_config_g0/1#switchport voice-vlan 400

Switch_config#voice-vlan mac-address 000b.82fe.932e mask ffff.ffff.ff00

NB: Here, Voice VLAN is 400 and MAC address *000b.82fe.932e* is for IP phone.

We may add a specific single MAC address or vendor based MAC address. First 6 degit of vendor MAC and rest of 6 digits will be zero. For example if Grandstream vendor's MAC addrss first 6 digit is **00:0b:82** then we may add the below command

Switch_config#**voice-vlan mac-address 000b.8200.0000 mask ffff.ff00.0000**

It's meaning all Grandstream's IP phone will be connected and vocie VLAN will be working perfectly.

51. How to configure Stacking or BVSS (BDCOM Virtual Switch System)

```
Swith#config
Switch_config#bvss
Switch_config_bvss#bvss enable
Switch_config_bvss#bvss domain-id 1
Switch_config_bvss#bvss member-id 1
Switch_config_bvss#bvss mode enhanced
Switch_config_bvss#bvss interface 1 type tGigaEthernet port 1 group 1
Switch_config_bvss#exit
Switch_config#exit
Switch#write bvss-config
```

Need to reboot the switch

S5828 (TG0/1) --- (TG0/1) S5828

Switch-2

```
Swith#config
Switch_config#bvss
Switch_config_bvss#bvss enable
Switch_config_bvss#bvss domain-id 1
Switch_config_bvss#bvss member-id 2
Switch_config_bvss#bvss mode enhanced
Switch_config_bvss#bvss interface 1 type tGigaEthernet port 1 group 2
Switch_config_bvss#exit
Switch_config#exit
Switch#write bvss-config
```

Need to reboot the switch

52. How to configure Break out cable in BDCOM 40G port. Or how to make 40/100G port as 10G port

Switch_config_cg0/1#switchport flex-portmode separate
Swith_config#exit
Switch#write vos-config
Switch#reboot

53. How to add Pallas in Switch/OLT

Switch_config#mqtt client server pallas.bdcom.cn
Switch-config#ip name-server 8.8.8.8
Swith_config#wr all

54. How to configure STP/RSTP in OLT PON port or in the ONU?

STP (Spanning Tree Protocol) is a layer 2 network protocol that helps prevent loops in Ethernet networks by blocking redundant paths. Typically, STP operates at the data link layer (Layer 2) of the OSI model and is used in Ethernet networks.

In the context of PON (Passive Optical Network) technology, such as in OLT (Optical Line Terminal), the PON port itself doesn't pass STP frames. PON technology operates at the physical layer and is point-to-multipoint, meaning that it's a one-to-many communication model. Each Optical Network Unit (ONU) communicates directly with the OLT over the PON, and there is no need for STP to handle redundant paths which we can see in a traditional Ethernet network.

In a PON environment, STP is typically not necessary because the topology is inherently different from a traditional Ethernet network. The design of PON networks eliminates the need for STP by avoiding loops in the physical topology.

55. Why packet discards in the switch port?

is not able to handle the incoming traffic volume.

- **Errors on the Line**: Physical issues such as faulty cables, connectors, or ports can cause errors that lead to discarded packets. CRC (Cyclic Redundancy Check) errors, for instance, occur when there's data corruption, and the switch discards the corrupted packets.
- **Port Misconfiguration**: Mismatches in configurations like speed, duplex settings, or VLAN assignments between the switch port and the connected device can lead to packet loss. For example, if one side is set to full duplex and the other to half duplex, collisions may occur, leading to discards.
- **Access Control Lists (ACLs)**: If there are ACLs or security policies applied to a port that filter certain types of traffic, the switch may drop packets that do not meet the allowed criteria.
- **Buffer Overflow**: A switch port has limited buffer memory. When the buffer overflows due to excessive traffic, the switch will discard additional incoming packets until space frees up.
- **QoS (Quality of Service)**: If QoS policies are in place, packets that exceed the allocated bandwidth or priority limits might be discarded to maintain the integrity of prioritized traffic.
- **Broadcast Storms or Loops**: A misconfigured network with loops or excessive broadcast traffic can overwhelm a switch, causing it to discard packets to protect its own resources.

56. How to configure RSPAN?

Switch_config#mirror session 1 destination interface g0/1 portbased rspan 10
Switch_config#mirror session 1 source interface g0/2 both

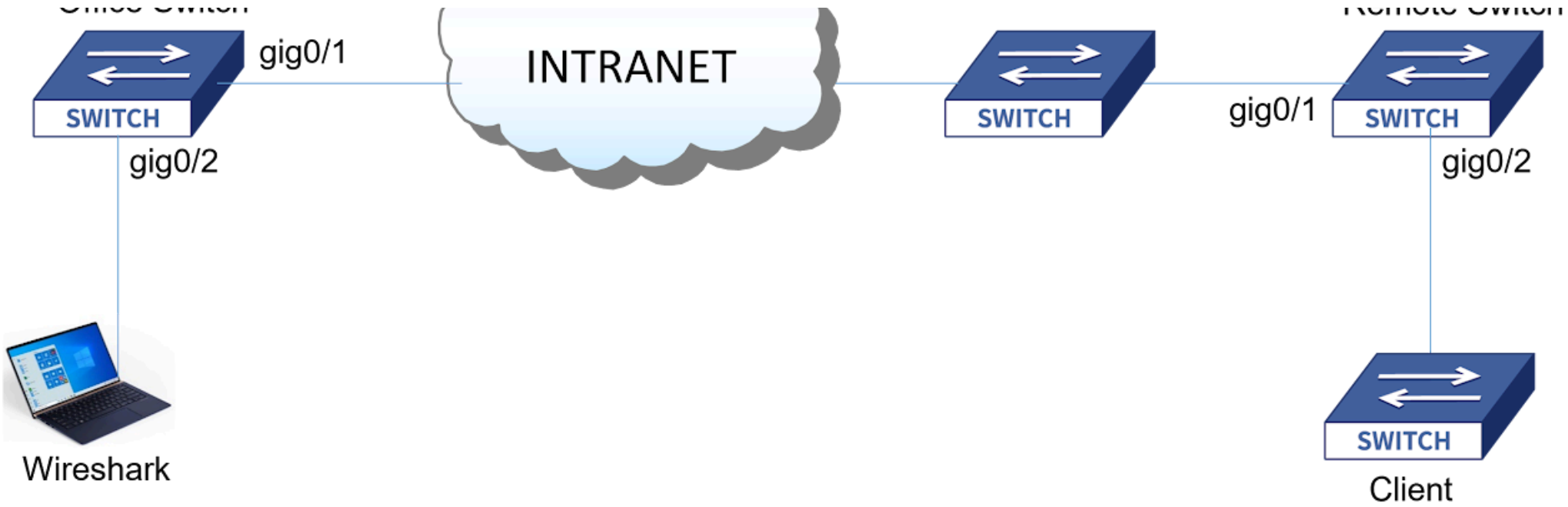
Switch_config#interface GigaEthernet0/1
Switch_config_g0/1#switchport mode trunk
Switch_config_g0/1#switchport trunk vlan-allowed 10

Office Switch:

Swith#config
Switch_config#mirror session 1 destination interface g0/2 portbased rspan 10
Switch_config#mirror session 1 source interface g0/1 both

Switch_config#interface GigaEthernet0/1
Switch_config_g0/1#switchport mode trunk
Switch_config_g0/1#switchport trunk vlan-allowed 10

Switch_config#interface GigaEthernet0/2
Switch_config_g0/2#switchport mode trunk
Switch_config_g0/2#switchport trunk vlan-allowed 10



NB: We have to create a VLAN and pass through that VLAN in every switch. Here we created VLAN **10**

57. How to configure HGU Ont TR-069 ACS Server?

Switch_config#gpon profile onu-flow-mapping *TR069-HGU*
Switch-gpon-profile# gpon-profile entry 1 uni type veip all
Switch-gpon-profile# gpon-profile entry 1 virtual-port 1
Switch-gpon-profile# gpon-profile entry 2 uni type ip-host 1
Switch-gpon-profile# gpon-profile entry 2 virtual-port 1
Switch-gpon-profile# exit

Step02: Make a VEIP profile mentioning ACS server information

Switch_config#gpon profile onu-veip *bdcom-acs*
Switch-gpon-profile#gpon-profile mgmt bind ip-host 1
Switch-gpon-profile#gpon-profile tr-069 url *http://tr-069.net*
Switch-gpon-profile#gpon-profile tr-069 username *test* password *bdcom*
Switch-gpon-profile#gpon-profile tr-069 vlan *10*

Step03: Apply the profile on the ONU

Switch_config#interface gpON 0/1:1
Switch_config_gpon0/1:1#gpon onu veip 1 veip-profile *bdcom-acs*
Switch_config_gpon0/1:1#gpon onu flow-mapping-profile *TR069-HGU*
Switch_config_gpon0/1:1#gpon onu ip-host 1 option dhcp

NB: Here the profile name is *bdcom-acs* the URL is *http://tr-069.net* username is *test* password is *bdcom* and the VLAN is *10*